

Jason Seh

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EDUCATION

B.S. in Computer Science & Engineering, Ohio State University

Expected May 2029

Member: Luma Consulting

WORK EXPERIENCES

The Ohio State University

Columbus, OH

Undergraduate Research Assistant | Python, PyTorch, DINOv2, SAM, OpenCV

April 2026 - Present

- Engineered an end-to-end agricultural computer vision pipeline using PyTorch, integrating foundational models like DINOv2 for zero-shot classification and Segment Anything (SAM) for automated leaf segmentation.
- Developed an automated, metadata-driven curriculum generation engine using Python, Pydantic, and Jinja2, mapping dataset complexity metrics to ABET-aligned learning outcomes.
- Processed large-scale image datasets using OpenCV and NumPy to calculate quantitative plant damage (HSV analysis), supporting experiential AI literacy for K-12 students.

PROJECTS

[OncoPath](#) | *Next.js, FastAPI, PyTorch, XGBoost, SHAP, Transformers (ViT), Three.js*

Feb 2026 - April 2026

- Engineered 21 individually trained XGBoost and PyTorch models using the MSK-MET dataset (25k+ records) to predict organ-specific metastatic risk across 21 anatomical sites with an average 0.72 AUC-ROC and multimodal capabilities.
- Fine-tuned a Phikon Vision Transformer (ViT) on 3TB of pathology imagery to extract 768-dimensional embeddings, fusing histopathology data with genomic profiles to enhance predictive accuracy for breast cancer subtypes.
- Architected a real-time FastAPI backend to execute asynchronous predictive simulations, integrating explainable AI to showcase the specific impact of high-impact drivers (e.g., KRAS, SMAD4, TP53).
- Developed an interactive 3D clinical dashboard using Next.js and React Three Fiber, featuring procedural anatomical heatmaps and Claude-powered prognostic predictions for 12-month treatment simulations supported through model results.

[BioForge](#) | *Python, Docker, AutoDock Vina, RDKit, Biopython, NumPy*

April 2026 - Present

- Architected an end-to-end drug discovery pipeline that automates the identification of high-affinity molecular binders for viral protein targets by bridging the gap between structural bioinformatics and physics-based docking simulations
- Engineered a Dockerized simulation wrapper using Python and AutoDock Vina to calculate binding affinity, validating the core by replicating the high-affinity benchmark of Imatinib for the BCR-ABL protein (1OPJ).
- Developed an automated structural preprocessing engine using Biopython and Gemmi to retrieve, clean, and refine 3D atomic coordinates from the PDB while identifying optimal active site pockets.
- Implemented a cheminformatics validation layer using RDKit to enforce Lipinski's Rule of Five and chemical sanitization, ensuring candidate molecules meet pharmacological standards for drug-likeness and stability

[Acelt - HackAI](#) | *Next.js, FastAPI, WebRTC, PyTorch, Librosa, Whisper, Gemini 3.0, Supabase*

Feb 2026

- Developed a high-performance, AI-driven mock interview platform providing real-time multimodal analysis (computer vision and audio) to evaluate candidate performance across behavioral and technical tracks. Placed 7th out of 100s of participants.
- Engineered an asynchronous audio intelligence using Whisper for transcription, Librosa for signal processing, and PyTorch for vocal metrics, optimizing the response time from 30s to <6s (80% reduction).
- Architected a scalable backend using aiohttp and aiortc for low-latency media processing and asynchronous biometric logging to Supabase, optimizing for concurrent user sessions.

[Buckeye Sense - MakeOhio](#) | *React, Vite, Supabase, Postgis, Deck.gl, MapLibre, ESP32*

March 2026

- Architected an asynchronous, end-to-end realtime pipeline that ingests high-frequency hashed Wi-Fi telemetry from a distributed ESP32 sensor network to visualize campus-wide human density in a 3D environment for fire rescue situations.
- Engineered a high-speed computational layer that automatically triangulates and plots device coordinates in milliseconds upon insertion into Supabase, utilizing PostGIS for sub-second location plotting.

TECHNICAL SKILLS

Languages: Python, TypeScript, JavaScript, Java, C#

Frameworks & Libraries: Next.js, React.js, Node.js, FastAPI, Aiohttp

Database: Supabase, PostgreSQL, MongoDB

Tools & Platforms: Docker, Cloudflare R2, Vercel, Git, Prisma

AI/ML: PyTorch, XGBoost, Transformers (ViT), SHAP, Scikit-Learn